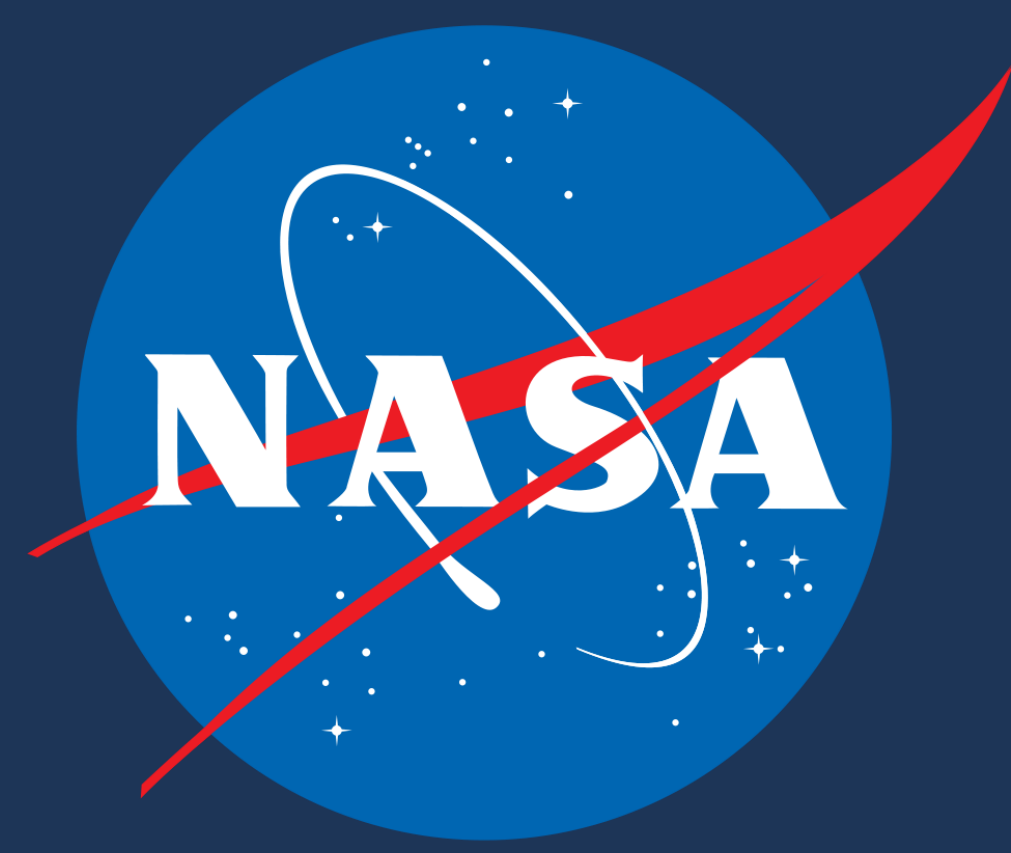




A Comparison Between Top-Down and Bottom-Up Methane Flux Calculations in the Great Dismal Swamp

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Background

Methane (CH₄)

- Accounts for 12% of all GHGs
- Approximately 80 times more effective as a GHG than carbon dioxide (CO₂)
- Wetlands account for 20 – 40% of all natural emissions of CH₄
- Wetlands cause high levels of uncertainty in climate models

The Great Dismal Swamp

- 452 km² wetland on the border between Virginia and North Carolina
- Methane fluxes studied from 1980 - 1981 and in 2018 using bottom-up techniques

Research Questions

- How does top-down (TD) analysis compare to bottom-up (BU) analysis?
- How has methane flux in the Great Dismal Swamp changed over time?

Methods

Data collection done through Picarro Cavity Ring-Down Spectrometer on NASA P-3 and Dynamic Aviation B200 during SARP 2025

Analysis techniques include top-down, estimate a point source from a plume, and bottom-up, estimating a plume from a point source

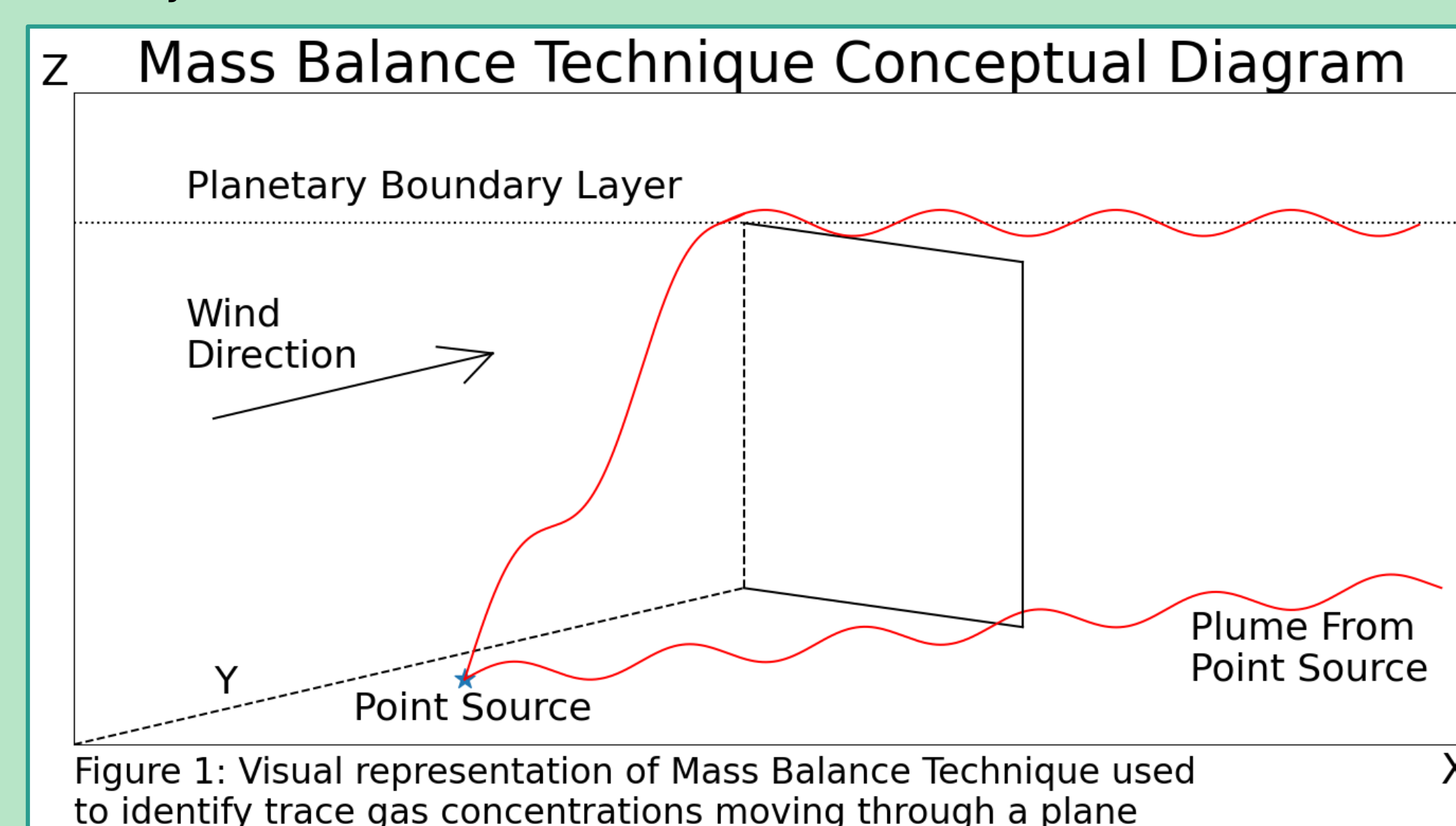
Analysis completed using Python to determine wind speed, wind direction, planetary boundary layer height, surface elevation, crosswind distance, methane concentrations, and verify natural methane sources.

Mass Balance Technique used to calculate methane flux of the Great Dismal Swamp (Fig. 1)

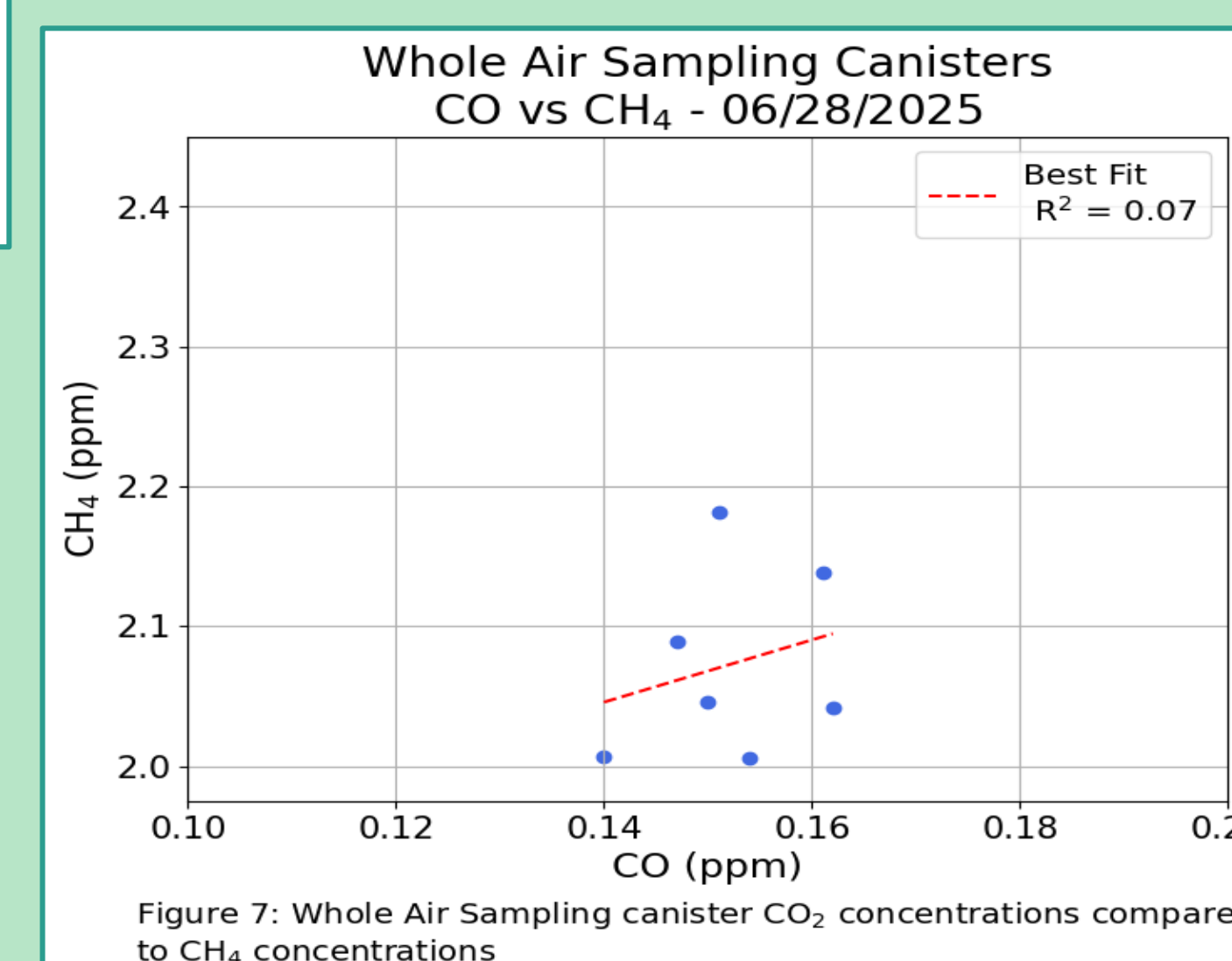
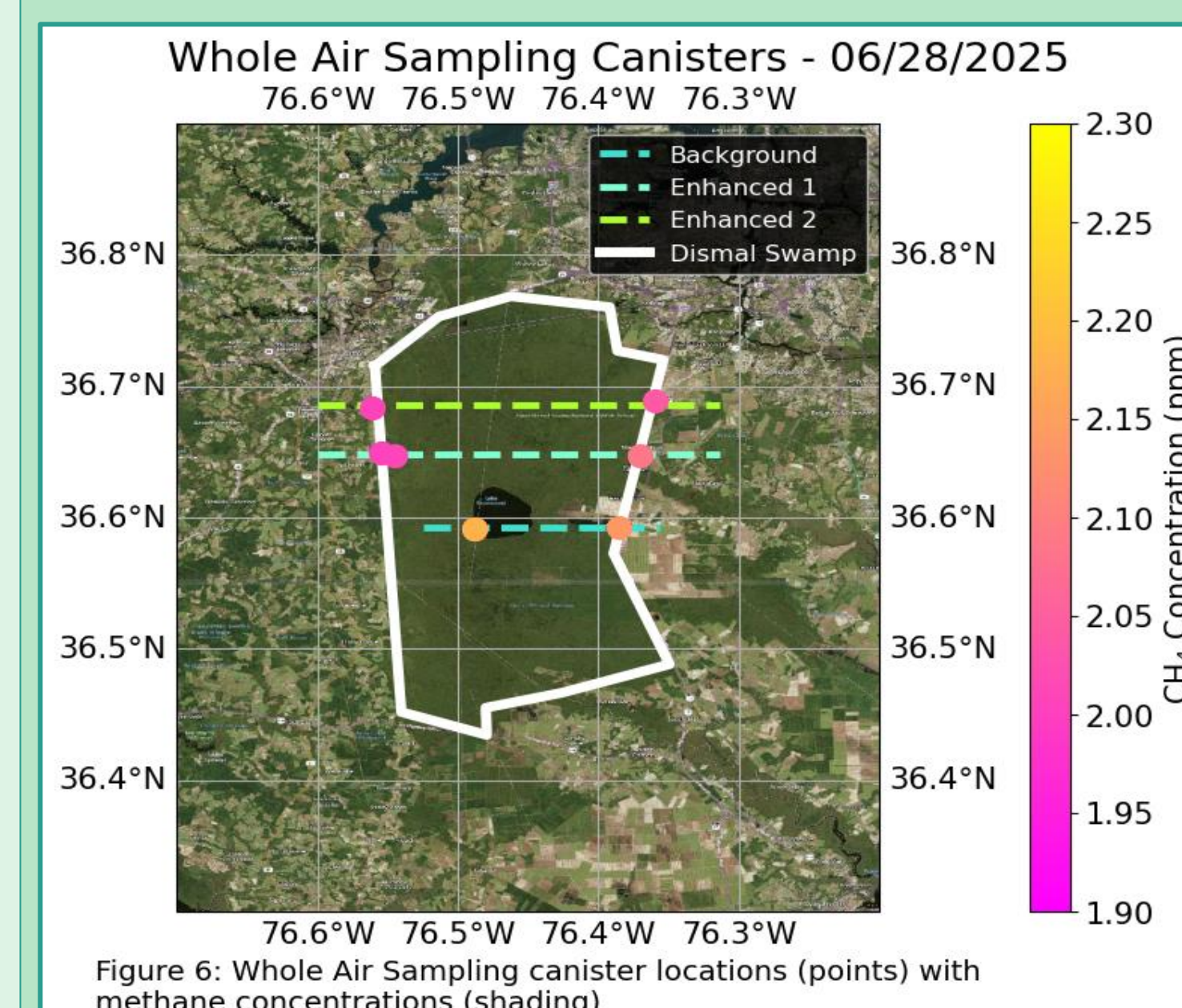
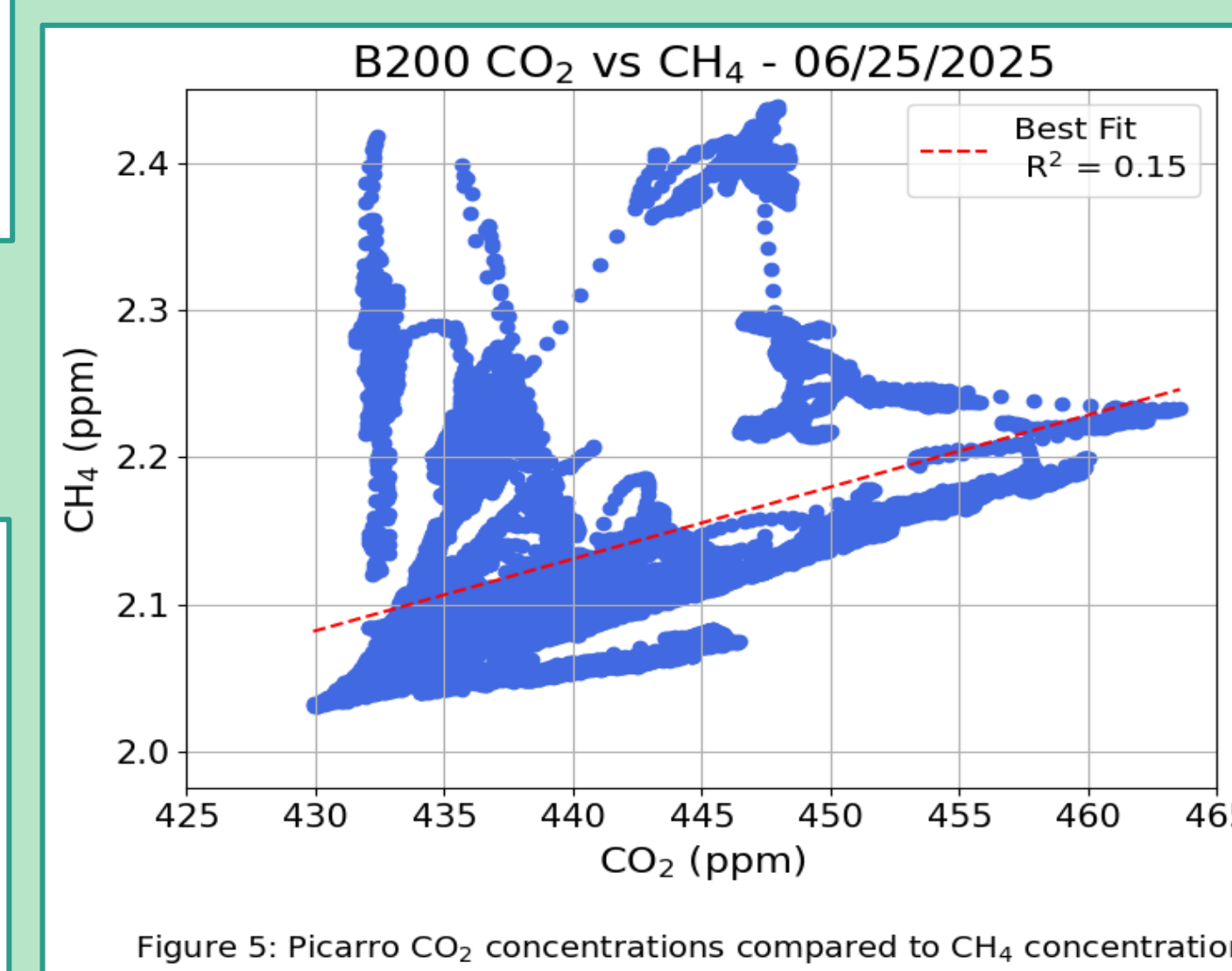
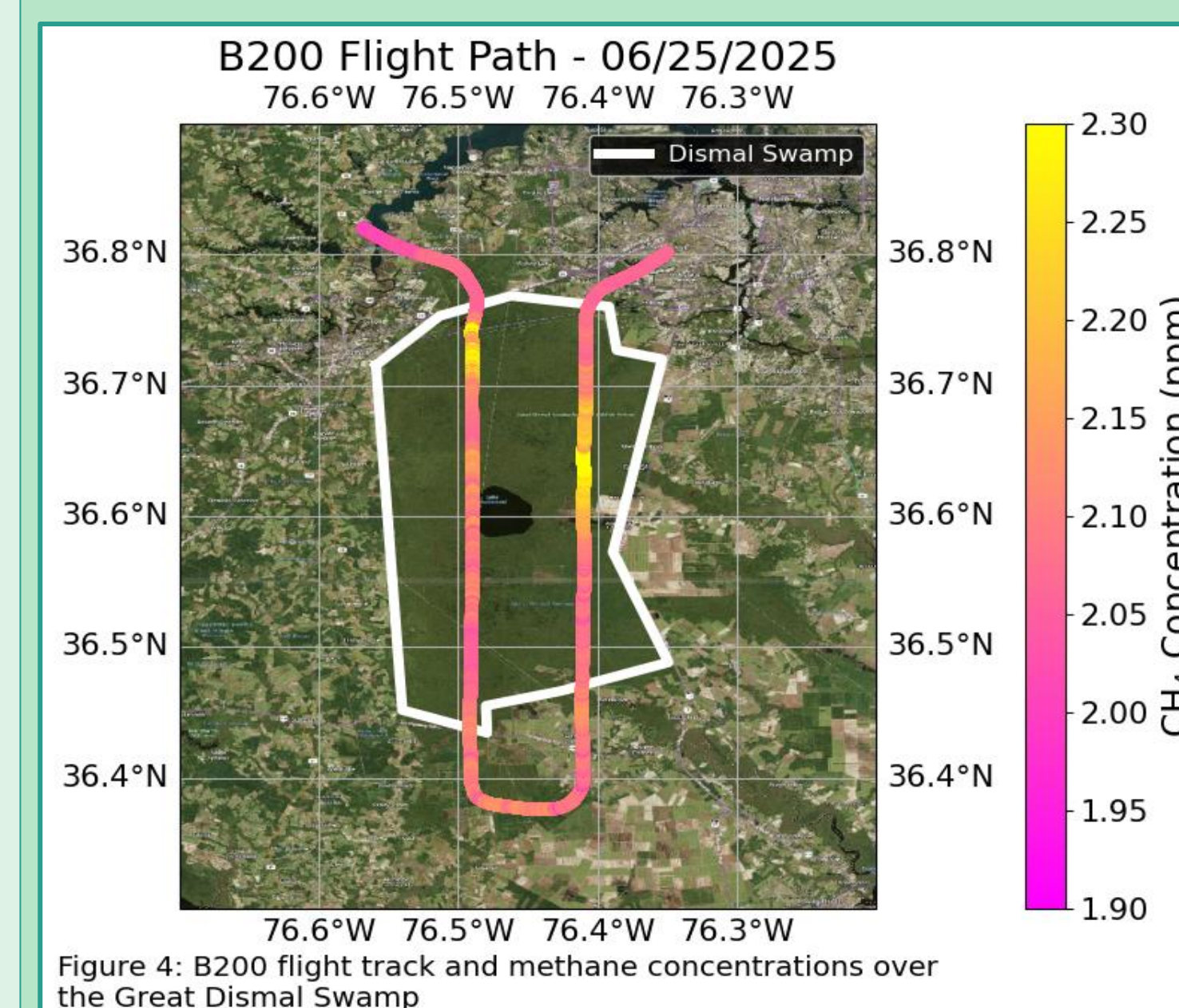
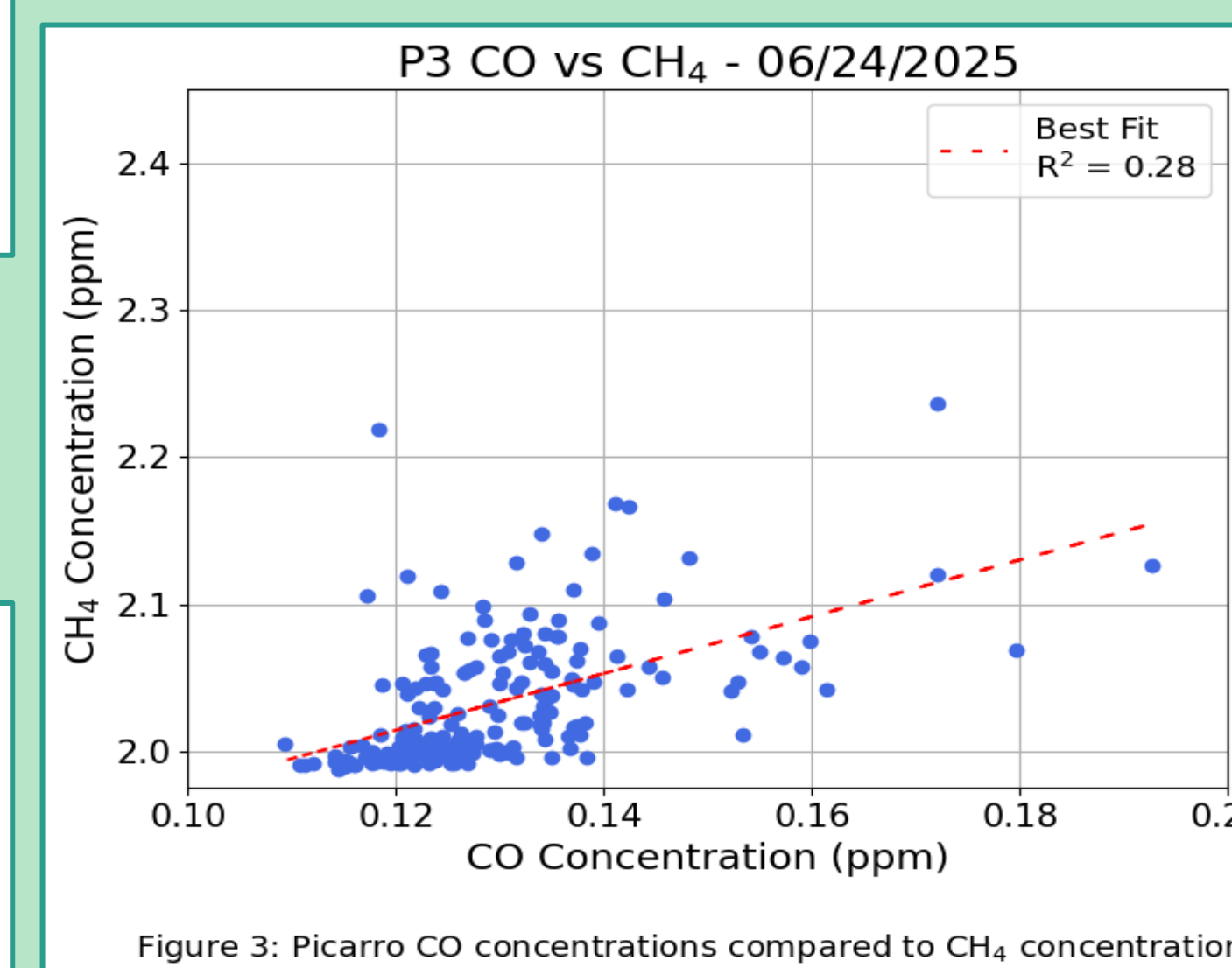
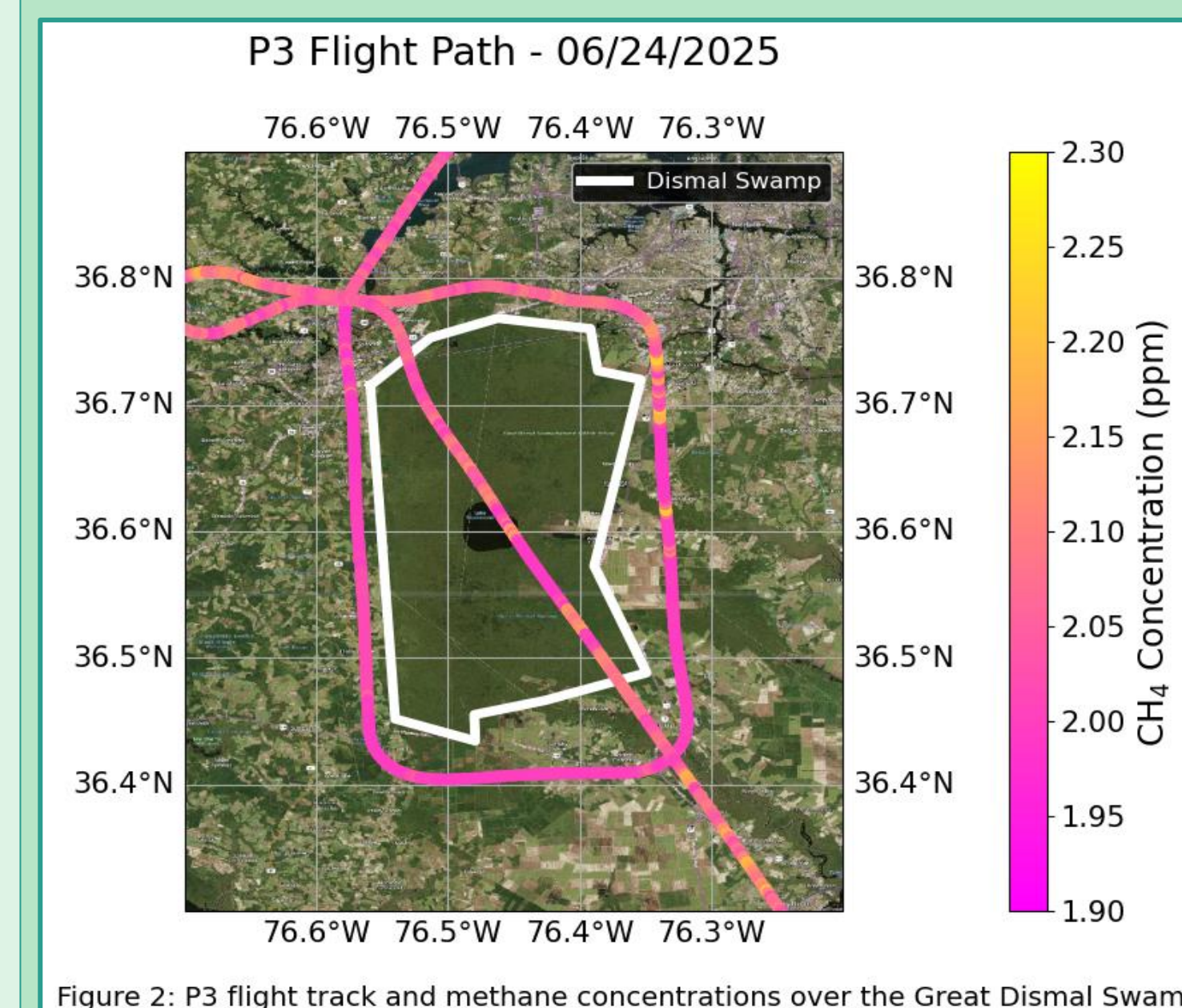
$$flux = v \cos(\alpha) \int_{z_0}^{z_1} \int_{-y}^y (X - X_{bg}) dy dz$$

Labels for the equation:

- Enhancement of Trace Gas (points to $X - X_{bg}$)
- Wind speed perpendicular to flight path (points to $v \cos(\alpha)$)
- Integral from surface to planetary boundary layer (points to $\int_{z_0}^{z_1}$)
- Integral over length of flight track (points to $\int_{-y}^y$)
- Background trace gas concentration (points to X_{bg})
- Enhanced trace gas concentration (points to X)



Methane Measurements & Correlations



Results & Key Findings

	Harris & Sebachner, 1982 (BU)	Gutenberg et. al., 2019 (BU)	WAS Sample (BU)	B200 25 th (TD)	B200 23 rd (TD)	P3 23 rd (TD)	P3 24 th (TD)	All Bottom Up	All Top Down
Average CH ₄ Flux (kg/s)	0.033	0.099	-0.181	-1.760	0.344	0.037	0.603	-0.559	0.140
Max CH ₄ Flux (kg/s)	0.103	0.099	-0.180	-1.760	0.854	0.037	0.603	0.103	0.854
Min CH ₄ Flux (kg/s)	0.006	0.099	-0.181	-1.760	-0.015	0.037	0.603	-1.181	-1.760

Top-down and bottom-up analysis return **similar** results

Bottom-up methods showed **lower levels** of methane flux compared to top-down methods

Variability on a day-to-day timescale

Discussion

SARP 2025 provides basis for comparing TD and BU methane fluxes but does not provide a complete picture.

Previous studies provide additional BU flux measurements but lack concurrent TD measurements which limits accurate comparisons.

This comparison shows that TD and BU flux measurements provide similar results.



Future Work

Collect more TD and BU measurements

Collect and compare simultaneous TD and BU measurements

Compare daily methane flux factors (soil moisture, soil temperature, water table height, oxygen content of soil) to identify possible causes of daily fluctuations

Compare calculated flux measurement with chemical transport models



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